

Information Technologies for Industrial Engineers

Database

Why?

Topic	Details
Data persistence	Data is not lost when the app is refreshed.
Data integrity	Stores data in a consistent format.
Data security	Defines and manages data access permissions.
Data scalability	Stays fast even with "large volumes" of data.

Two types of database

- SQL
 - Table-based (like Excel)
 - Relational
- NoSQL
 - Document-based (document, key-value, or wide-column stores)
 - Non-relational

"SQL" Database

- PostgreSQL
- MySQL
- SQLite
- Microsoft SQL Server
- MariaDB
- Oracle

"NoSQL" Database

- MongoDB
- Redis
- Elasticsearch
- Dynamodb
- Firebase Cloud Firestore (ຈຳໄດ້ປາວ)

Popularity

<https://survey.stackoverflow.co/2025/technology#1-databases>

Let's create your own cloud database

<https://supabase.com/>

Tasks

- Sign up
- Create organization
- Create project
- Create table
- Input some data

SQL commands

Create table

```
CREATE TABLE IF NOT EXISTS
  todos (
    id serial PRIMARY KEY,
    title VARCHAR(50) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
  );
```

Delete table

```
DROP TABLE IF EXISTS todos;
```

View data

```
SELECT * from todos;
```

```
SELECT  
  id,  
  title,  
  created_at  
from  
  todos;
```

Create data

```
INSERT INTO
  todos (title)
VALUES
  ('First Todo'),
  ('Second Todo'),
  ('Third todo');
```

```
INSERT INTO
  todos (id, title)
VALUES
  (2, 'First Todo'),
  (3, 'Second Todo'),
  (4, 'Third todo');
```

Delete rows

```
DELETE FROM  
  todos  
WHERE  
  id = 1;
```

```
DELETE FROM  
  todos  
WHERE  
  title = 'First Todo';
```

Let's make a todo app (Vue + supabase)

Getting info

- Get `Project ID`
 - "Project Settings" -> "General"
- `URL`
 - "Project Settings" -> "Data API"
- `API_KEY` (*anon, public*)
 - "Project Settings" -> "API"

Setting up

- `pnpm create vite@latest`
 - ...
- `pnpm install @supabase/supabase-js`
- `pnpm install -D supabase`

Setting up

- `pnpm exec supabase login`
- `pnpm exec supabase gen types typescript --project-id "PROJECT_ID" --schema public > ./src/types.ts`

```
./env.local
```

```
VITE_SUPABASE_URL=<SUBSTITUTE_SUPABASE_URL>  
VITE_SUPABASE_PUBLISHABLE_KEY=<SUBSTITUTE_SUPABASE_PUBLISHABLE_KEY>
```

The url should end with `.supabase.co` not `.supabase.co/rest/v1`

./src/db.ts

```
import { createClient } from "@supabase/supabase-js";
import type { Database } from "../types";
const supabaseUrl = import.meta.env.VITE_SUPABASE_URL;
const supabasePublishableKey = import.meta.env.VITE_SUPABASE_PUBLISHABLE_KEY;
export const supabase = createClient<Database>(
  supabaseUrl,
  supabasePublishableKey,
);
```

Viewing todos

https://github.com/it-for-ie-69/vue-db-todo/blob/main/src/App_todo_view.vue

Adding / deleting todos

https://github.com/it-for-ie-69/vue-db-todo/blob/main/src/App_todo_add_delete.vue

Full CRUD todo

https://github.com/it-for-ie-69/vue-db-todo/blob/main/src/App_full_crud.vue